Can Machine Learning Predict If Your TikTok Will Be a Hit or a Flop?

Predicting the Number of Likes of a Video from Its Metadata

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Machine learning project
github.com/NathanTessot75/tiktok-analysis

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Abstract

This study aims to predict the number of likes a TikTok video will receive from its characteristics and from metadata about its author. Using a dataset of 19,084 TikTok videos, I engineer 22 additional features and compare three regression models: a linear regression, a LASSO regression and a neural network (multilayer perceptron), with hyperparameters tuned by 5-fold cross-validation. Models are evaluated with the root mean squared error (RMSE) on a held-out test set. The neural network performs best, with a test RMSE of 44,533 likes and a test R^2 of 0.89, against about 50,000 likes and $R^2 \approx 0.86$ for the two linear models. Train and test errors are almost identical for all three models, which points to bias rather than overfitting as their main limitation.

1 Introduction

Predicting the number of likes a TikTok video will receive is a valuable task in social media analytics: it can guide content creation, advertising strategies and platform optimization. The goal of this project is to predict the number of likes of a video from its characteristics (duration, views, shares, comments, downloads, transcription) and from the status of its author (verified account, banned or under review).

Since the target is a continuous count, this is a regression problem. I compare a simple linear baseline, a regularized linear model and a non-linear model, to see how much flexibility the task requires.

2 Data

The data is a public dataset of TikTok videos, available on Kaggle, in which each video is labelled as stating a *claim* or an *opinion*. It contains 19,382 videos; removing rows with missing values leaves $N = 19{,}084$ videos. Each video is described by 12 variables: claim status, duration, transcription text, verified status and ban status of the author, and the counts of views, likes, shares, downloads and comments.

The number of likes is extremely skewed (Figure 1): the median video receives about 3,400 likes, while the mean is about 84,000 and the maximum exceeds 650,000. The distribution is clearly bimodal, which largely reflects the claim status: the median "claim" video receives about 124,000 likes, against about 800 for "opinion" videos.

3 Method

Feature engineering. I created 22 new variables, for a total of 27 predictors: engagement ratios (shares, comments and downloads per view; views per comment and per download; comments

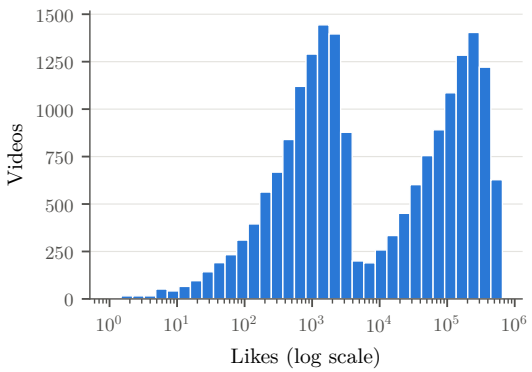


Figure 1: Distribution of the number of likes ($N = 19{,}084$ videos, log scale).

and shares per download); log transformations of the views, comments, shares and downloads, and a total engagement volume; duration-based rates (views, comments, shares and interactions per second); encoded categorical variables (verified account, ban status, claim status); and simple text features from the transcription (length, number of words, average word length).

Protocol. All predictors are standardized. The data is split into a training set (75%, 14,313 videos) and a test set (25%, 4,771 videos). Three models are trained:

- a **linear regression**, as a baseline;
- a **LASSO regression**, whose regularization strength α is tuned by 5-fold cross-validation over seven values from 1 to 10^6 (35 fits);
- a **neural network** (multilayer perceptron with early stopping), whose architecture, activation function, solver, learning rate and batch size are tuned by 5-fold cross-validation over 120 combinations (600 fits; Table 2).

Metric. Since I want to avoid large prediction errors, models are compared with the root mean squared error (RMSE), which penalizes large deviations more heavily. I also report the coefficient of determination R^2 on the test set.

4 Results

Table 1 summarizes the performance of the three models, and Figure 2 compares their predictions with the actual number of likes.

Model	Train RMSE	Test RMSE	Test R^2
Linear regression	49,436	49,927	0.864
LASSO regression ($\alpha = 1000$)	49,882	50,410	0.861
Neural network	**44,611**	**44,533**	**0.891**

Table 1: Performance of the three models (RMSE in number of likes).

4.1 Linear regression

The linear regression reaches an RMSE of 49,436 on the training set and 49,927 on the test set (Figure 2a). The small gap between the two suggests low variance, but the level of the error indicates that the model suffers from bias: a linear combination of the features cannot capture the whole relationship. The model also handles the many videos with few likes poorly: about a fifth of its test predictions are negative, which is impossible for a count.

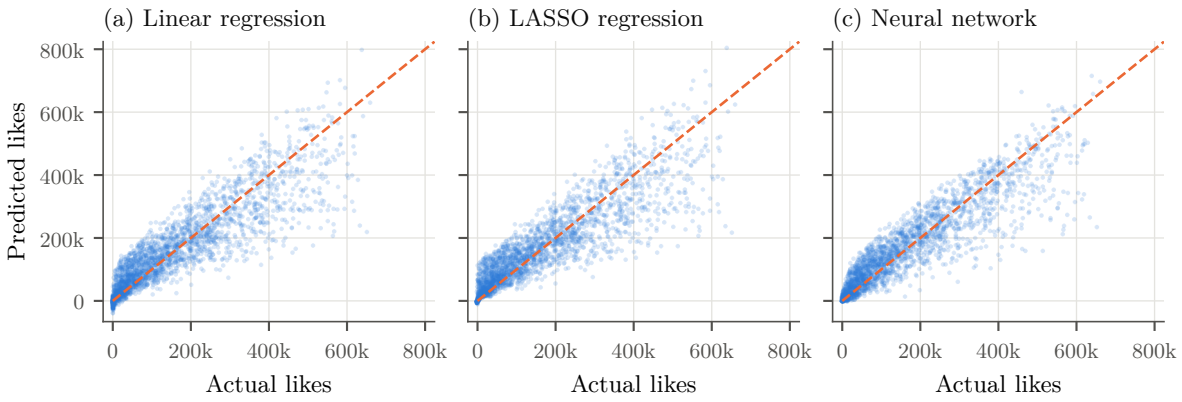


Figure 2: Predicted versus actual number of likes on the test set. The dashed line marks perfect predictions.

4.2 LASSO regression

Cross-validation selects $\alpha = 1000$. The LASSO reaches an RMSE of 49,882 on the training set and 50,410 on the test set (Figure 2b), very close to the unregularized model. Since the linear regression did not overfit, regularization brings no gain here: the problem is bias, not variance.

4.3 Neural network

The selected network has two hidden layers of 8 and 4 units with ReLU activations, trained with Adam (learning rate 0.01, batch size 32; Table 2). It reaches an RMSE of 44,611 on the training set and 44,533 on the test set (Figure 2c), about 11% lower than the linear models. Its predictions follow the diagonal more closely and are never negative: for videos with fewer than 10,000 likes, its mean absolute error is about 1,400 likes, against 4,800 for the linear regression.

The learning curve (Figure 3) shows that the network needs about 250 iterations to stabilize. Train and test errors stay very close throughout training and plateau together around 43,500 likes: the model does not overfit, and its remaining error is mostly bias.

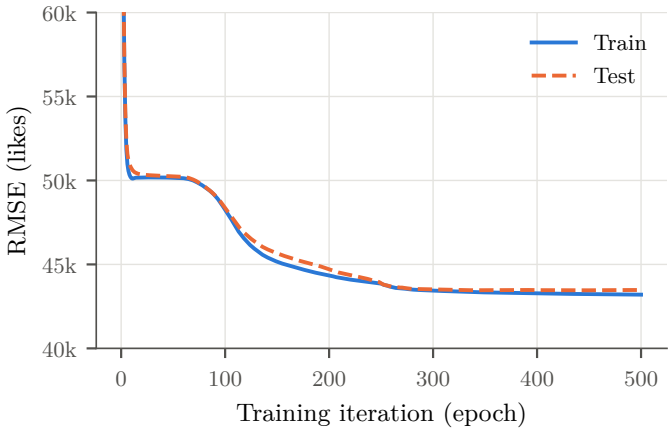


Figure 3: Train and test RMSE of the selected neural network across training iterations.

5 Conclusion

The neural network is the best model, with a test RMSE of 44,533 likes and a test R^2 of 0.89. Its error is about a third of the standard deviation of likes (about 135,000 on the test set), and

the close match between train and test errors shows that it generalizes well. User metadata and video characteristics therefore carry a strong signal about the number of likes.

These results come with two caveats. First, the RMSE is dominated by the most popular videos: an average error of 44,000 likes is small for a video with 500,000 likes but very large for the typical video, which receives about 3,400. Modelling the logarithm of likes, or reporting a relative error, would give a fairer picture across the whole distribution. Second, the strongest predictors (views, shares, comments) are measured at the same time as the likes. The models therefore explain the likes of a video that has already been published rather than predict, before publication, whether it will be a hit. Predicting success ahead of time would require features available at upload, such as the content of the video, its author's history or the publication time.

A Neural network hyperparameters

Hyperparameter	Tested values	Selected value
Hidden layer sizes	(1), (2), (2, 2), (4, 2), (8, 4)	(8, 4)
Activation function	ReLU, tanh, logistic	ReLU
Solver	Adam, SGD	Adam
Initial learning rate	0.001, 0.01	0.01
Batch size	32, 64	32

Table 2: Hyperparameters tested in 5-fold cross-validation (120 combinations, 600 fits). Early stopping is used in every configuration.